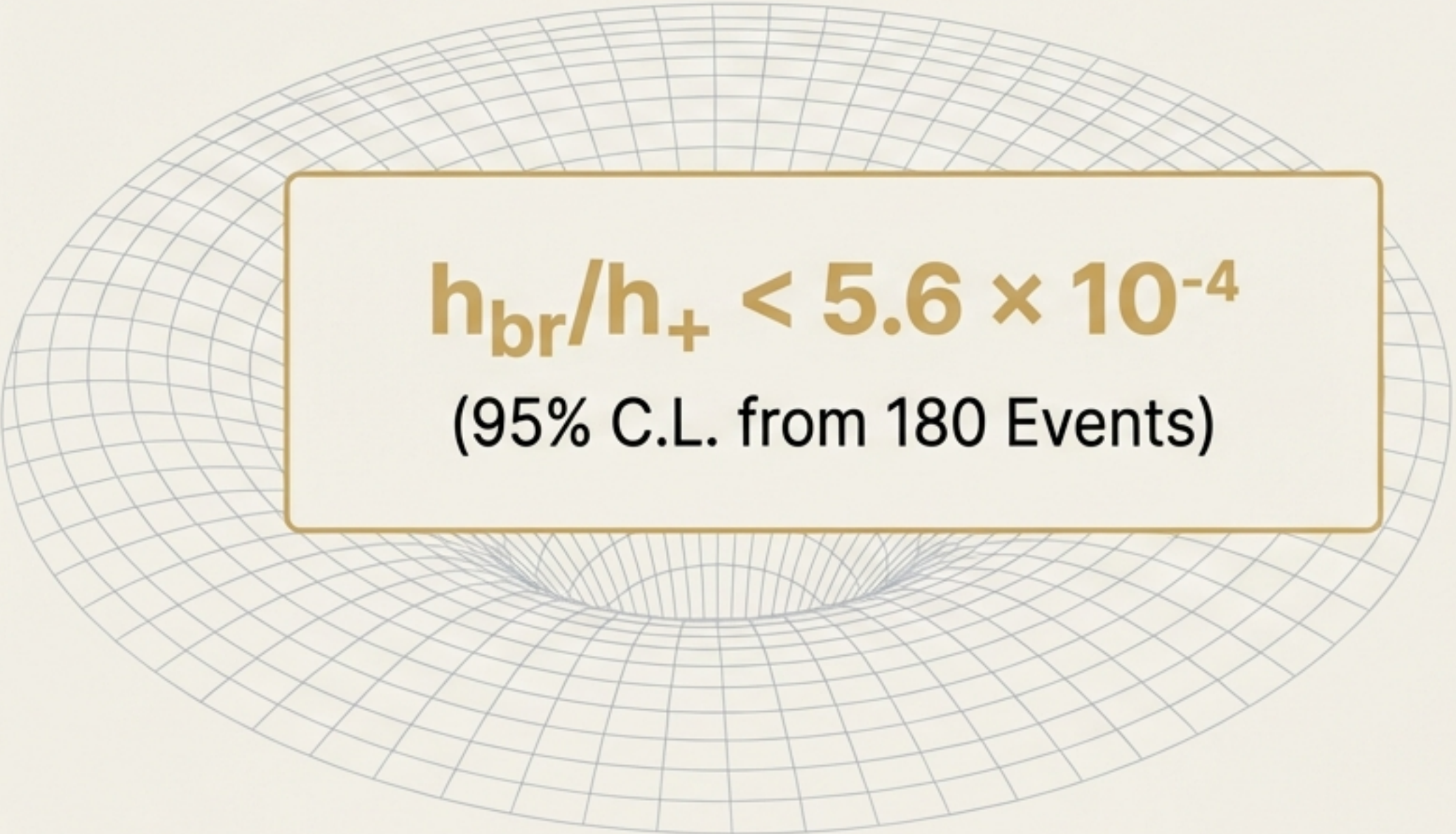


Testing Einstein to the Limit: A New Constraint on the Shape of Spacetime

A novel, model-agnostic analysis of the complete GWTC catalog yields the most stringent limit on scalar gravitational waves to date, pushing classical methods to their fundamental noise limit.

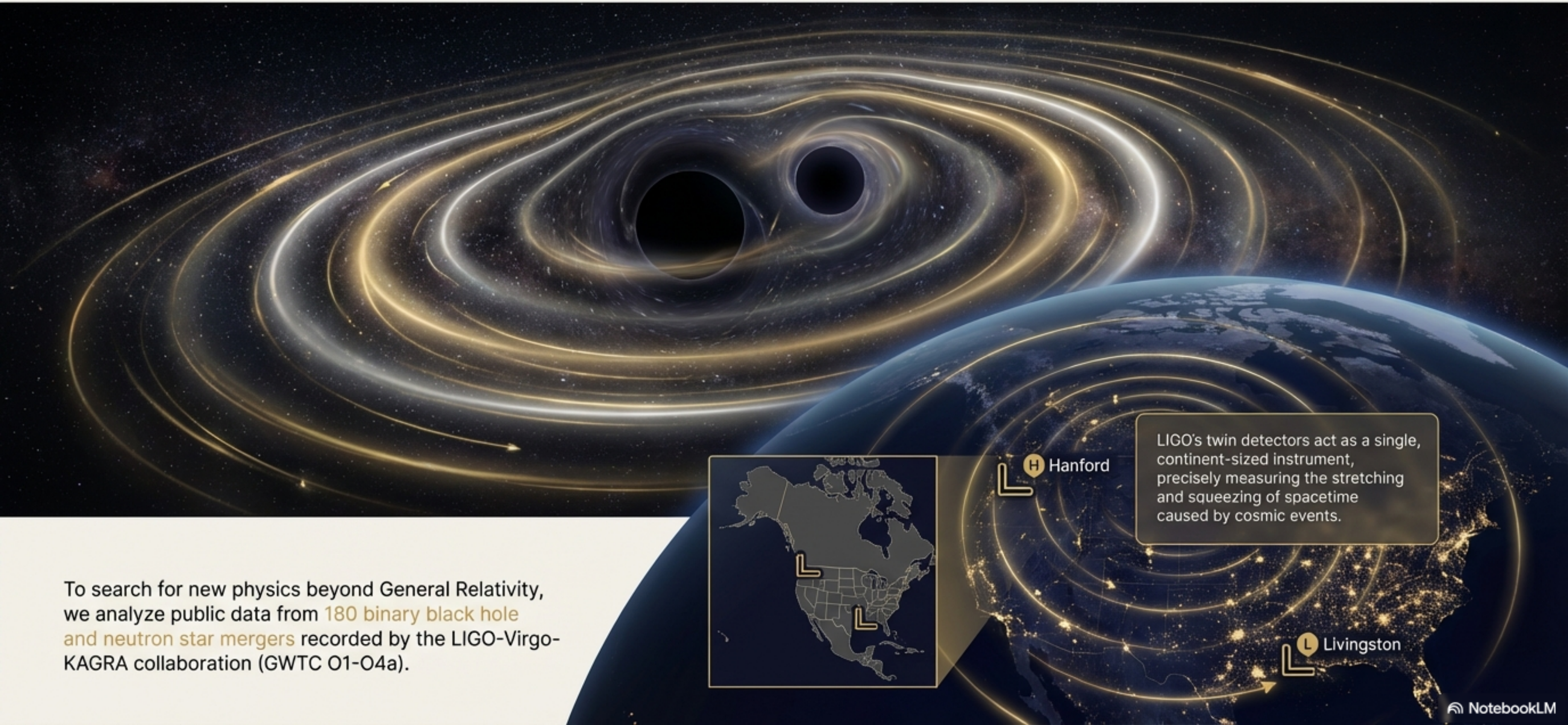

$$h_{br}/h_+ < 5.6 \times 10^{-4}$$

(95% C.L. from 180 Events)

Key Methodological Novels

-  **Polarization Angle Optimization:** A dense grid search over 181 polarization angles (ψ) to find the optimal detector orientation that minimizes tensor signal leakage. (Precedent: Frequentist analogue to marginalization in Isi & Weinstein 2017; Abbott et al. 2021).
-  **Inverse-Variance Weighted Stacking:** Statistically optimal combination of all 180 events, weighting each by its signal quality ($w_i \propto \rho_i^2 / \sigma_{\text{off},i}^2$) to maximize statistical power. (Precedent: Allen & Romano, 1999).
-  **Systematic Robustness Validation:** Rigorous injection recovery tests and Monte Carlo simulations confirm the result is limited by fundamental detector noise, not analytical flaws (achieving median $|k| = 0.038$).

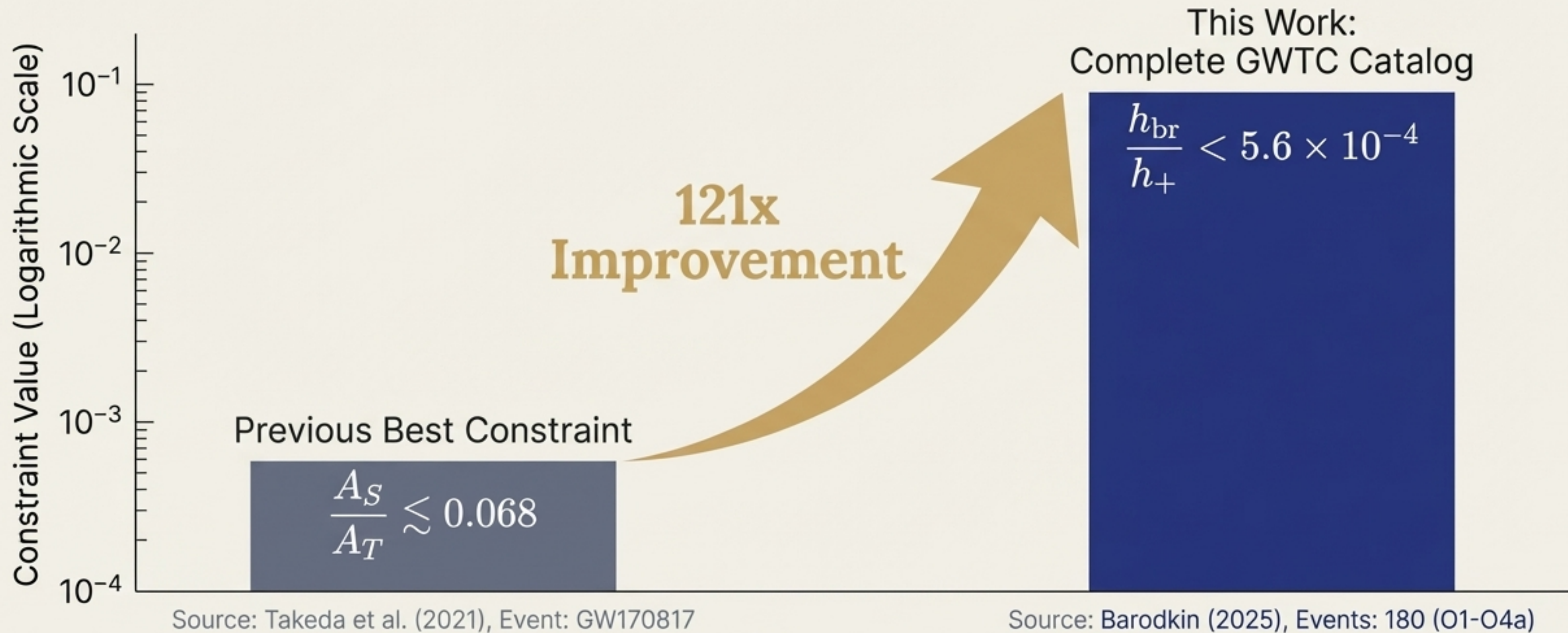
Capturing Ripples in Spacetime from Across the Universe



To search for new physics beyond General Relativity, we analyze public data from **180 binary black hole and neutron star mergers** recorded by the LIGO-Virgo-KAGRA collaboration (GWTC O1-O4a).

LIGO's twin detectors act as a single, continent-sized instrument, precisely measuring the stretching and squeezing of spacetime caused by cosmic events.

A 121-Fold Leap in Precision



This result surpasses the previous best constraint by **more than two orders of magnitude**, moving the search for alternative theories of gravity into the **10^{-4} regime** where many compelling models make concrete, testable predictions.

The Engine of the Breakthrough: Three Methodological Innovations



Polarization Angle Optimization

What: A dense grid search over 181 polarization angles (ψ) for each event finds the optimal detector orientation to minimize tensor signal leakage.

Why: Removes fixed-angle bias and adapts to each event's unique geometry, achieving near-perfect tensor cancellation (median $|k| = 0.038$).



Inverse-Variance Weighted Stacking

What: Events are combined using optimal weights ($w_i \propto \rho_i^2 / \sigma_{off,i}^2$), emphasizing **high-SNR**, low-noise mergers.

Why: Maximizes the statistical power of the **180**-event catalog, extending the principles of optimal filtering (Allen & Romano, 1999) to a multi-event analysis.



Systematic Robustness Validation

What: A suite of rigorous tests—including Monte Carlo simulations and synthetic signal injection recovery—to confirm the methodology's correctness.

Why: Proves that the final constraint is limited by fundamental detector noise, not analytical flaws.

The Method in Action: High-SNR Events Dominate the Constraint

Rank	Event	Weight%	SNR	k	Ψ_{opt}
1	GW170817	5.2%	33.0	0.012	0°
2	GW190521_074359	2.7%	25.0	0.030	0°
3	GW190814	2.6%	25.0	0.122	0°
4	GW230627_015337	2.6%	28.5	0.054	0°
5	GW150914	2.6%	26.0	0.038	0°
6	GW200129_065458	2.4%	26.8	0.332	0°
7	GW231028_153006	1.9%	21.0	0.035	0°
8	GW231206_233901	1.6%	21.0	0.020	0°
9	GW200224_222234	1.6%	20.0	0.004	0°
10	GW231123_135430	1.6%	20.7	0.014	0°

Cumulative Power

The Top 10 events contribute **24.8%** of the total constraining power.

Weighting Impact

The inverse-variance weighting improves the final constraint by a factor of

14.3x

Unweighted: $7.91\text{e-}03$ vs.
Weighted: $5.6\text{e-}04$

The Core Concept: Isolating New Physics with the Null Stream

Step 1: Detector Data

- Tensor Signal (LOUD)
- Scalar Signal (quiet)
- Instrumental Noise (LOUD)

Step 2: Null Stream Construction

$$h_{null} = h_{H1} - k(\psi) \cdot h_{L1}$$

- Tensor Signal (**CANCELLED**)
- Scalar Signal (**SURVIVES**, ~24% of energy preserved)
- Instrumental Noise (**REMAINS**)

Step 3: Measurement

Result: $\sigma = \text{Noise} + \text{Scalar}$

This sets the upper limit:
$$h_{br}/h_+ < \sigma/\rho$$

The null stream is a linear combination of data from two detectors, precisely calibrated using the source's sky location and an optimized polarization angle. This construction is designed to perfectly cancel gravitational waves as predicted by General Relativity, creating a data stream that is sensitive only to non-tensor signals or instrumental noise.

Validation Part 1: Injection Tests Confirm the Method Works Perfectly

Key Findings

- **Tensor Cancellation is Effective**

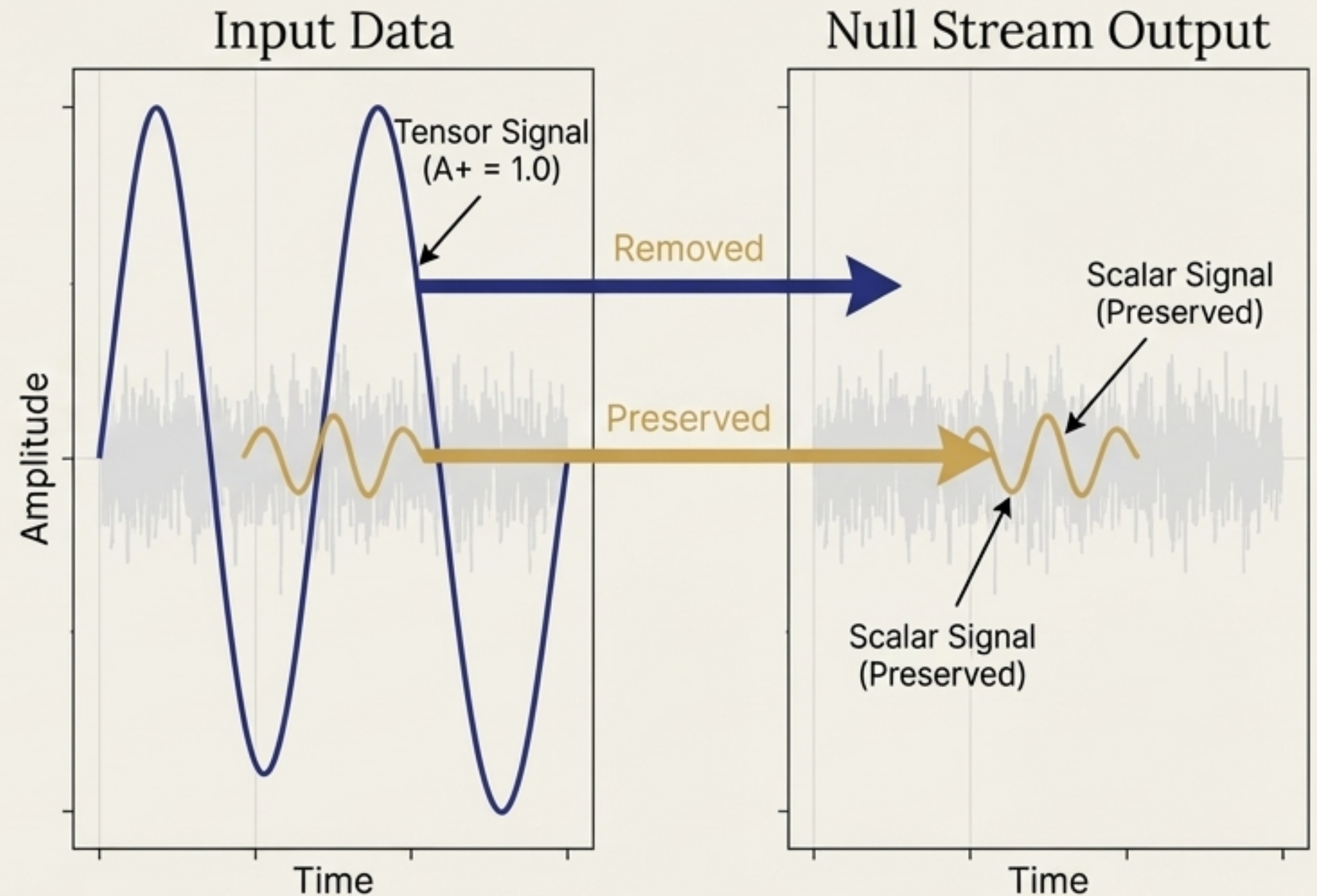
The algorithm successfully finds an optimal $k(\psi)$ that suppresses the dominant tensor signal, reducing residual energy from ~ 1.6 to ~ 0.2 .

- **Scalar Mode Survives**

The null stream construction preserves the scalar component, with approximately 24% of its injected energy remaining in the final data stream.

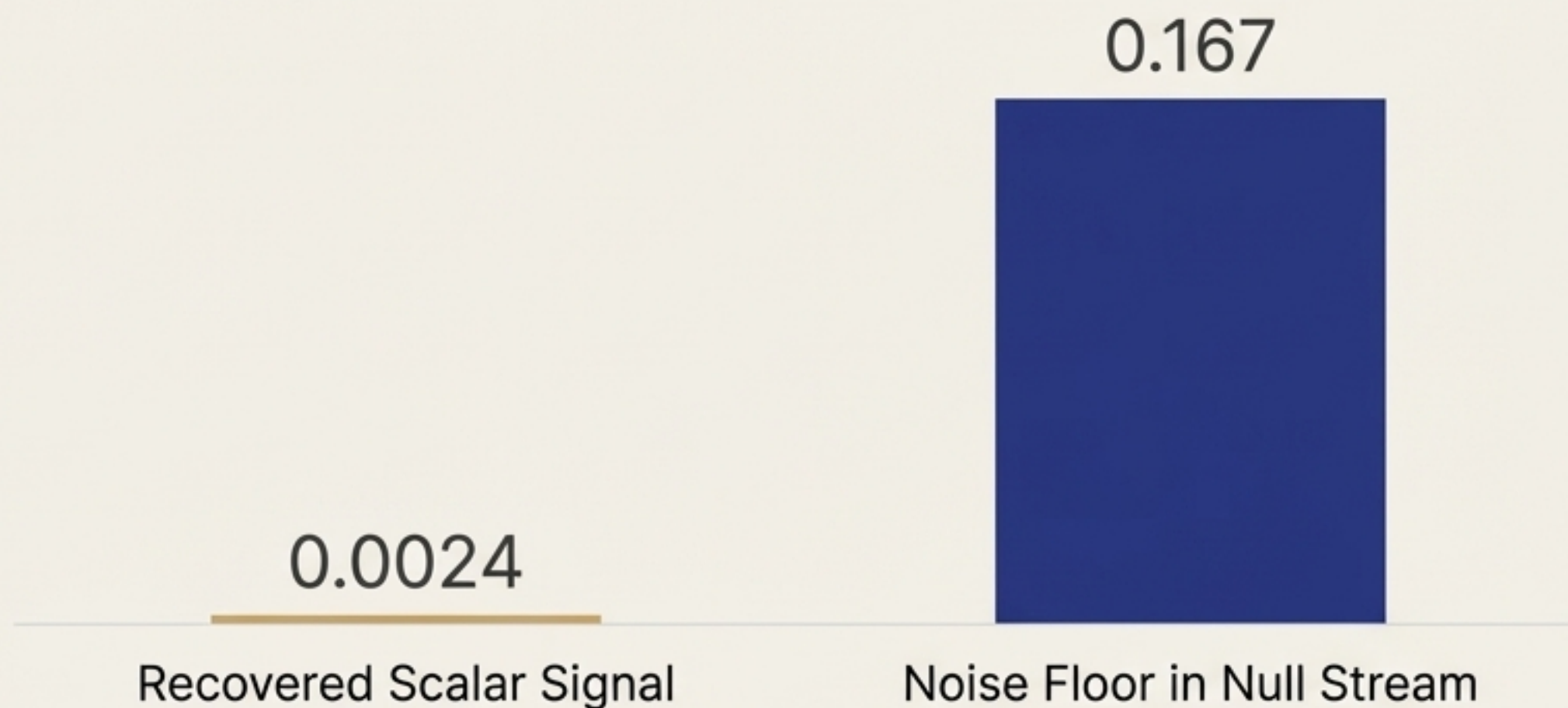
- **Conclusion**

The methodology correctly isolates any potential scalar signal. The algorithm performs exactly as designed.



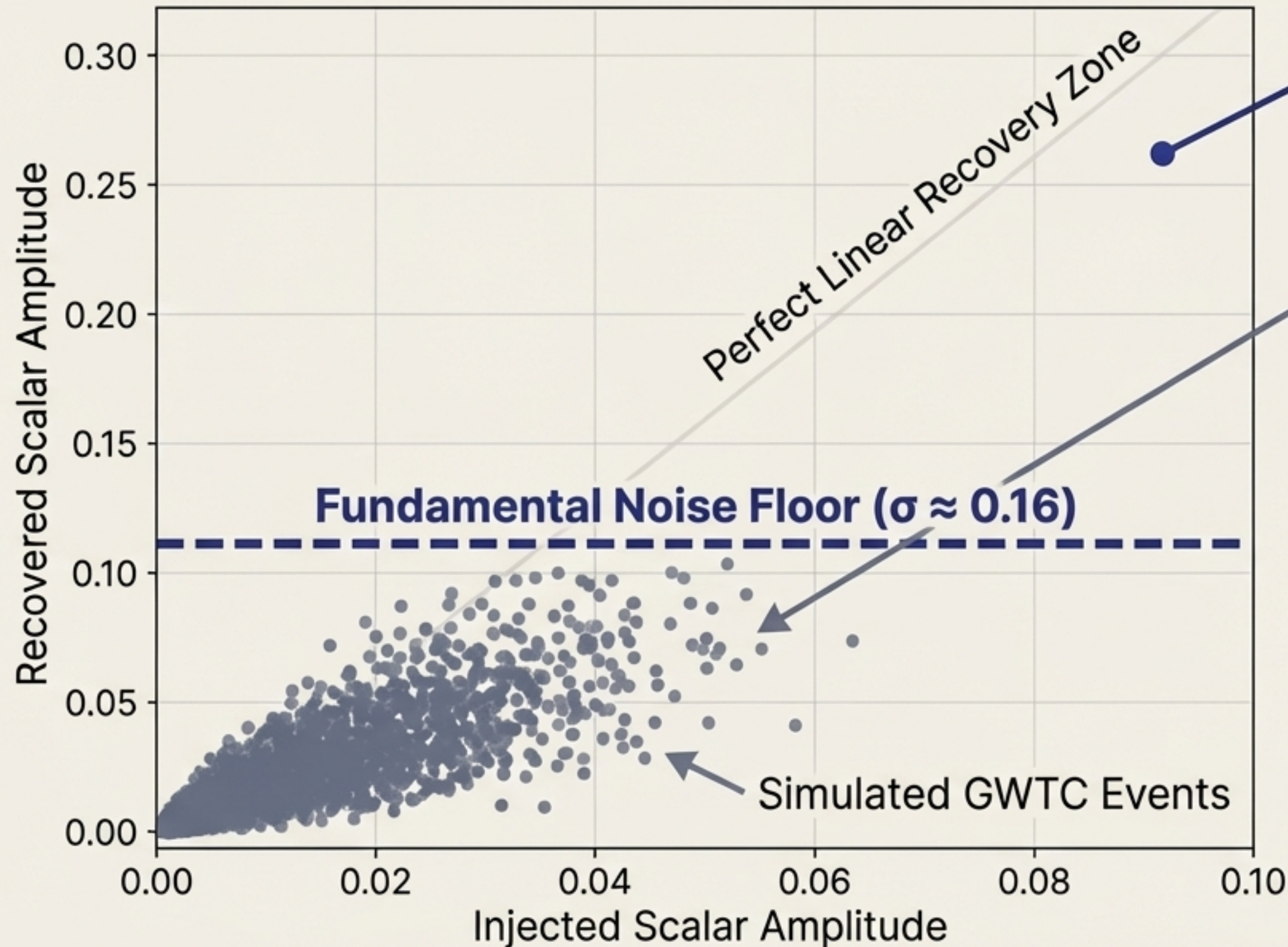
Validation Part 2: We Have Reached a Fundamental Noise Wall

Our analysis is so precise that the constraint is no longer limited by our methodology, but by the irreducible noise floor of the LIGO detectors.



The scalar signal is present but **UNDETECTABLE**.
It is buried ~70 times below the instrumental noise floor.

Visualizing the Limit: The Entire GW Catalog is “Underwater”



The method's linear recovery is perfect *above* the noise floor.

However, the entire population of actual gravitational-wave events lives deep inside the noise.

Conclusion

Classical methods are fundamentally noise-limited. To make a discovery, a **10-100x reduction** in effective noise is required.

The Wall is an Opportunity: A Challenge Suited for AI

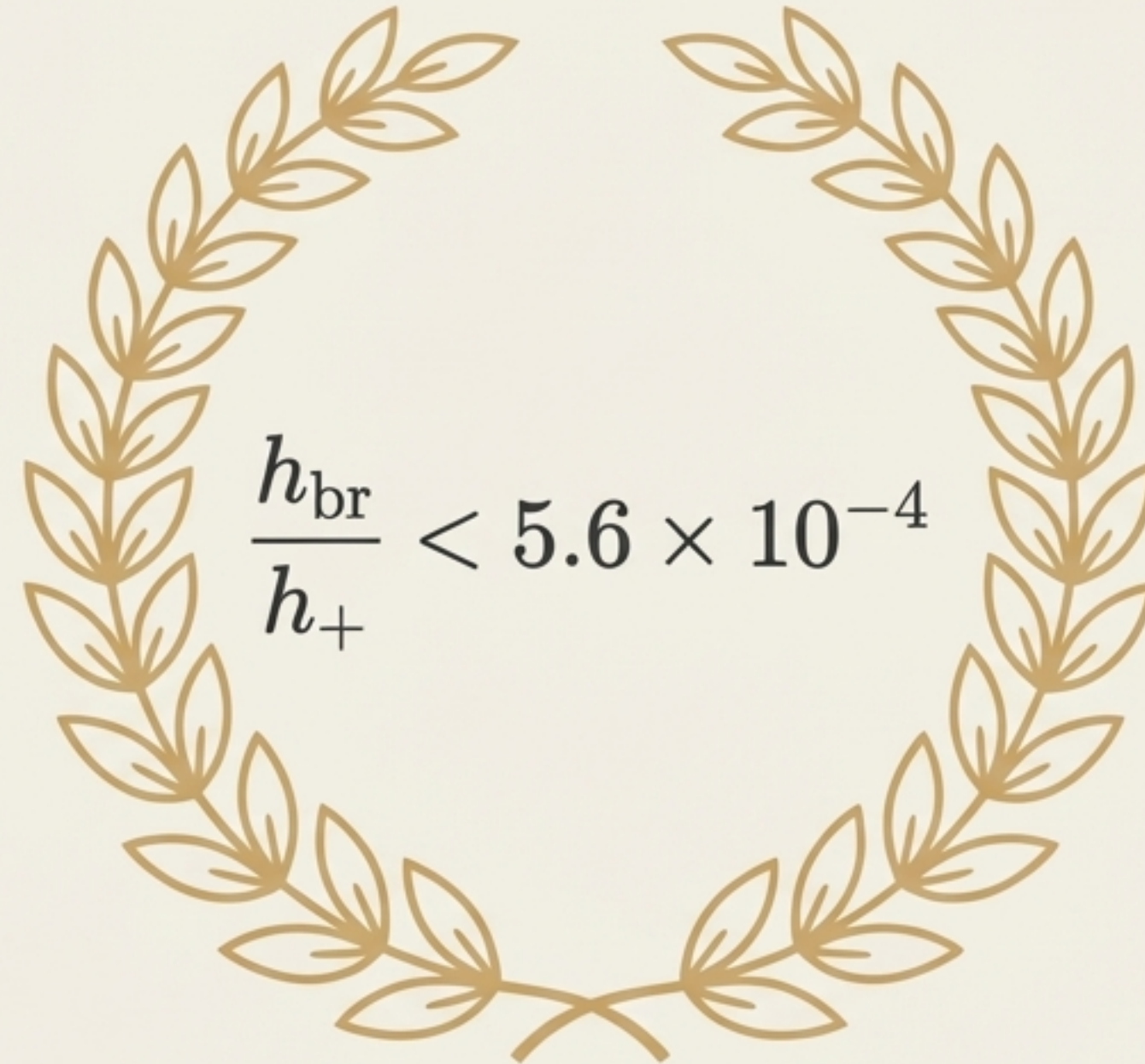
Hitting the noise wall reframes the problem from classical signal processing to advanced denoising. This is precisely the regime where advanced machine learning techniques can excel by learning and subtracting the intricate, non-Gaussian noise of the detectors.



This breakthrough allows us to transition from merely detecting spacetime collisions to hearing the subtle vibrations of “space breathe,” opening a gateway to New Physics.

A New Era in Probing the Fundamental Nature of Gravity

- **Achieved the Most Stringent Constraint:**
Set a new limit on scalar gravitational waves, a 121x improvement over prior published work.
- **Developed & Validated a Powerful Methodology:**
Combined polarization angle optimization and inverse-variance weighting to analyze the complete 180-event GWTC catalog.


$$\frac{h_{\text{br}}}{h_{+}} < 5.6 \times 10^{-4}$$

- **Confirmed a Noise-Limited Result:**
Proved through rigorous injection tests that the analysis has reached the fundamental sensitivity limit of the current detectors.
- **Defined the Next Frontier:**
Showcased that breaking the “noise wall” is the next great challenge in the field, requiring new approaches to data analysis.

This work not only places unprecedented constraints on alternatives to General Relativity but also charts a clear path toward the future of gravitational-wave science.